Rajshahi University of Engineering & Technology

Department of Computer Science and Engineering

Course Code : CSE 3106
Course Title : Computer Interfacing and Embedded Systems Sessional

Lab Report - 02

Name of the Experiment: Experimenting the Implementation of TMP36 Temperature Sensor and Photoresistor using STM32 Blue Pill.

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Title:

Experimenting the Implementation of TMP36 Temperature Sensor and Photoresistor using STM32 Blue Pill.

Introduction:

The TMP36 temperature sensor is a low-voltage, precision centigrade sensor that outputs a voltage that is proportionate to the Celsius temperature. It is recognised for its simplicity, low cost, and great accuracy, making it suitable for a wide range of applications, including consumer electronics and scientific research. The TMP36 achieves typical accuracies of ±1°C at ambient temperature. Its temperature is ranged between -50°C to 125°C and voltage output is ranged between 0V to 1.75V.

Photosensors sense the intensity of light and are essential components of light-sensitive devices. These sensors transform light energy into electrical impulses that can be monitored and analysed. Photosensors have a wide range of applications, including autonomous lighting systems, photography, and plant growth monitoring.

Experiment Procedure:

- **Creating the Environment:** At first we installed different softwares to compile and upload our codes to the STM32 Blue Pill board. Some of the softwares were: Arduino IDE/Keil IDE, STM32CubeProgrammer, STSW Link etc.

- **TMP36 Temperature Sensor:**
  - Components Needed: STM32 Blue Pill, TMP36 Temperature Sensor, RGB LED Light, Breadboard, Resistors, Connecting Wires
  - At first we coded the necessary logics and functions in our IDE and then compiled our code. After successful compilation, we move to our bread board connection.
  - The GND rail on the breadboard is connected to the GND pin on the STM32 Blue Pill.
  - The 5V rail on the breadboard is connected to the 5V pin on the STM32 Blue Pill.
  - TMP36 Connection:
    - Pin 1 (left pin): Connected to the 5V pin on the STM32 Blue Pill.
    - Pin 2 (middle pin): Connected to Analog pin A0 on the STM32 Blue Pill.
    - Pin 3 (right pin): Connected to Ground (GND) on the STM32 Blue Pill .
  - RGB LED Connection:
    - Common Pin (longest pin, cathode): Connected to Ground (GND) on the breadboard.
    - Red Channel Pin: Connected to a resistor and then to Digital Pin B2 on the STM32 Blue Pill.
    - Green Channel Pin: Connected to a resistor and then to Digital Pin B1 on the STM32 Blue Pill.
    - Blue Channel Pin: Connected to a resistor and then to Digital Pin B0 on the STM32 Blue Pill.

- **Photoresistor:**
  - Components Needed: STM32 Blue Pill, Photosensor, LED Light, Breadboard, Resistors, Connecting Wires
  - At first we coded the necessary logics and functions in our IDE and then compiled our code. After successful compilation, we move to our bread board connection.
  - The GND rail on the breadboard is connected to the GND pin on the STM32 Blue Pill.

- The 5V rail on the breadboard is connected to the 5V pin on the STM32 Blue Pill.
- LED Connection:
  - Anode (longer leg) of the LED: Connected to a resistor and then to Digital Pin PA0 on the STM32 Blue Pill.
  - Cathode (shorter leg) of the LED: Connected to the Ground (GND) rail on the breadboard.
- Photoresistor Connection:
  - One leg of the photoresistor is connected to 5V on the STM32 Blue Pill.
  - Other leg of the photoresistor is connected to a 10kΩ resistor, which is then connected to Ground (GND) on the breadboard (Black wire). Also connected to Analog Pin A1 on the STM32 Blue Pilll.